Ex. No.:5	AUTOMATED LEGAL DOCUMENT ANALYZER USING GPT-4	URK23CS1167
09.10.2026		
GitHub Link: https://github.com/MONICK10/EXP_5_GPT4		

Aim:

To develop an automated legal document analyzer using GPT-4 that summarizes legal documents and extracts important information such as parties, key clauses, obligations, payment terms, termination conditions and important dates.

Problem Statement:

Design and develop an automated legal document analyzer using GPT-4. The application should accept legal contracts or legal text, analyze the content, generate a concise summary, identify important clauses and obligations, extract relevant information and present the results in a structured format.

Software Requirements:

Software	Version
Python	3.13.5
Ollama	0.40.13
VS Code	1.144.0
Pandas	3.1.0
Sentence Transformers	6.2.0
Flask	3.1.3
LLM Model	Llama 3.2 (3B) via Ollama

Input Parameters:

Parameter	Description
Legal Document	Contract or legal text – uploaded .txt/.pdf/.docx file or pasted text. Example: employment_agreement.txt
Document Type	Auto-detect, Employment, NDA, Rental, Service, Sales, Loan or Other
Parties	Parties mentioned in the document (extracted automatically)
Document Length	Number of characters, words and approximate pages (calculated automatically)
Analysis Type	Complete, Summary, Clauses or Obligations
Output Format	Structured text report with headings and bullet points

System Architecture:

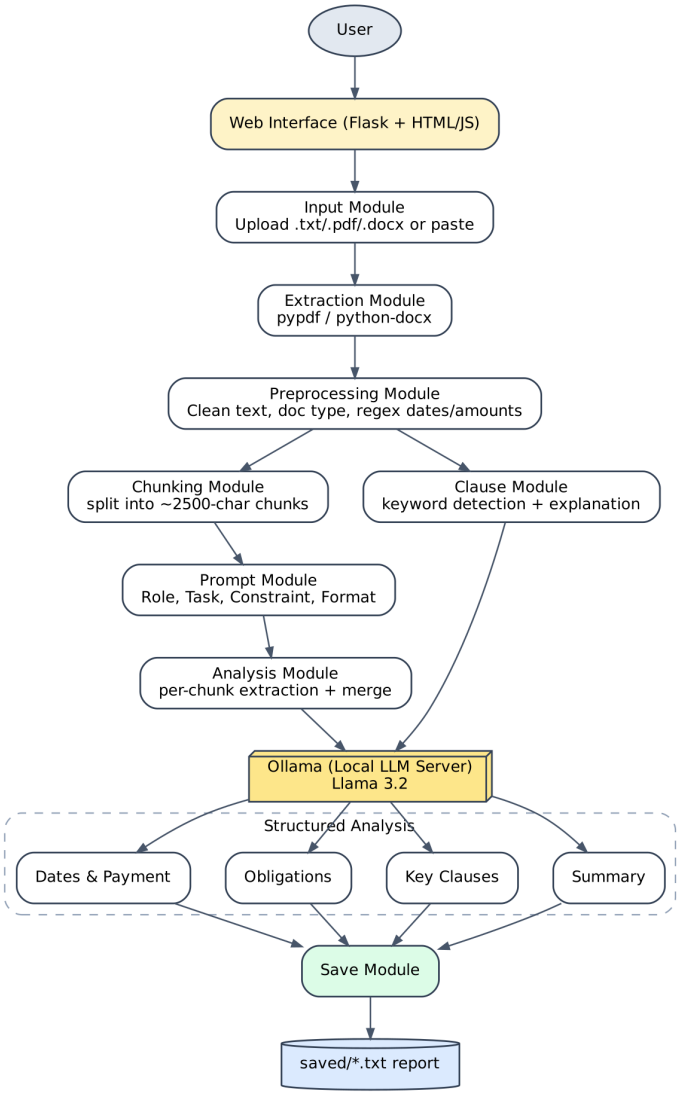


Fig. 5.1 System architecture

Module Design:

Module	Responsibility
Input Module	Accepts legal documents through file upload (.txt, .pdf, .docx), sample selection or manual text entry.
Extraction Module	Reads the uploaded document and extracts its text using pypdf and python-docx.
Preprocessing Module	Cleans the text, detects the document type and finds dates, amounts and durations using regular expressions.
Chunking Module	Splits long documents into chunks of about 2500 characters at paragraph boundaries.
Prompt Module	Creates a structured prompt with Role, Task, Information to extract, Constraint ('do not invent facts') and JSON Output Format.

Module	Responsibility
Analysis Module	Sends each chunk to the LLM, merges the extracted parties, obligations, payment, termination and dates, and generates an abstractive summary.
Clause Module	Identifies key clauses (confidentiality, payment, termination, liability, IP, dispute resolution) and explains each in simple words with importance.
Save Module	Saves the complete structured analysis report as a .txt file.

Implementation:

- GitHub Link - https://github.com/MONICK10/EXP_5_GPT4
- LinkedIn Post Link - <add LinkedIn post link in config.json>
- Run: `python app.py` → `http://localhost:5005`

Output Screenshots:

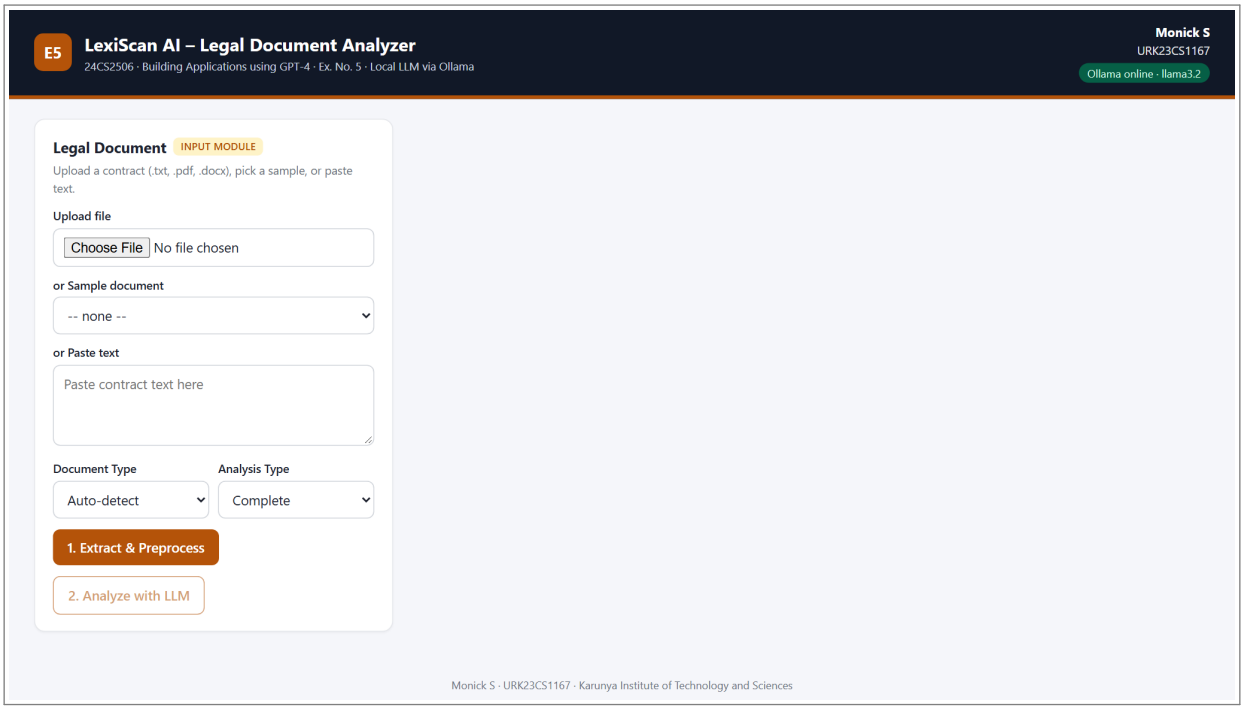


Fig. 5.2 Home Screen

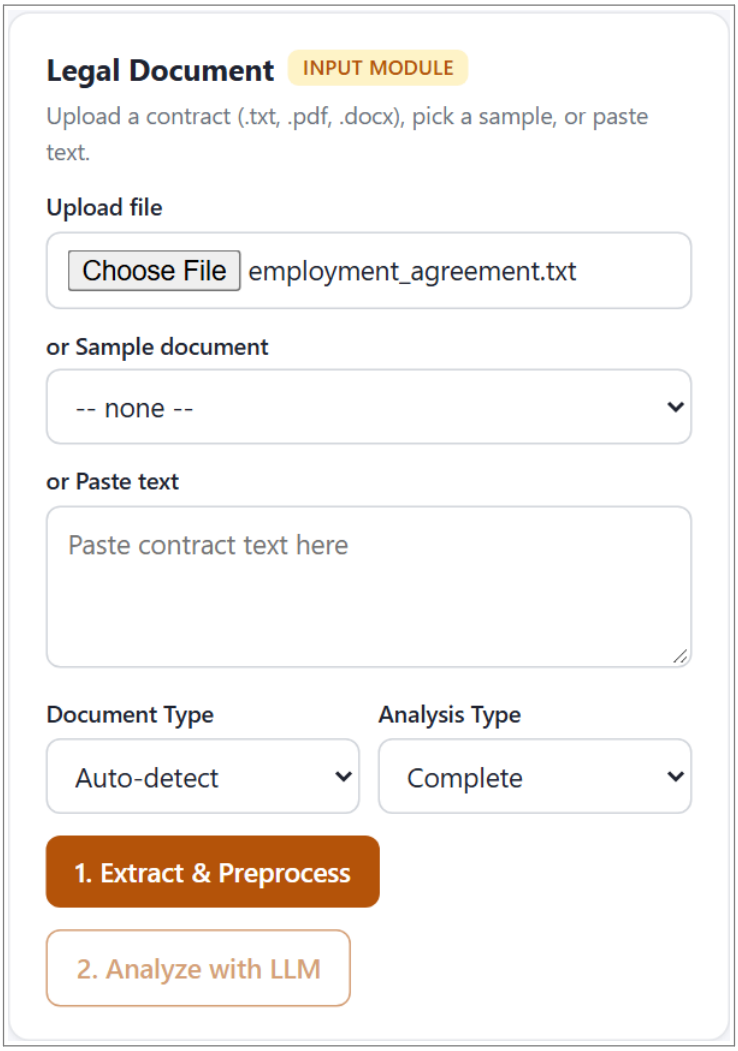


Fig. 5.3 Document Upload

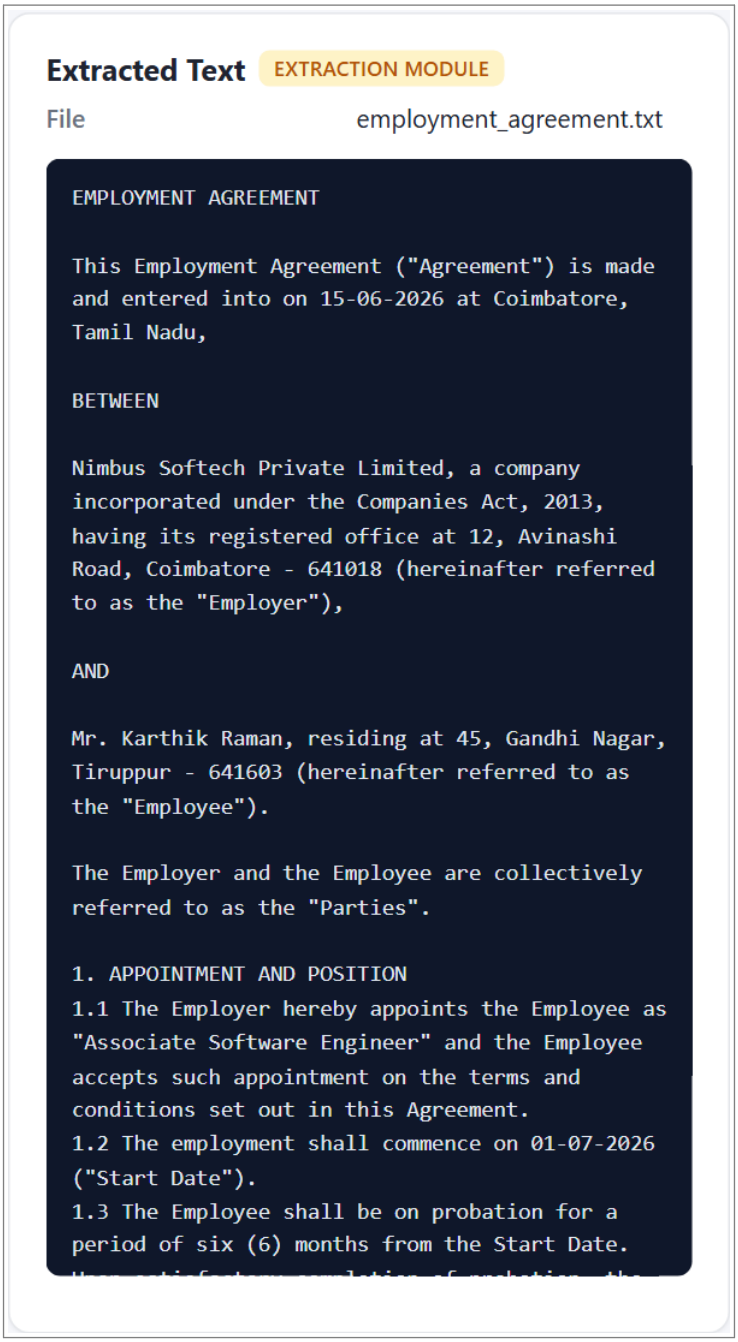


Fig. 5.4 Extracted Text

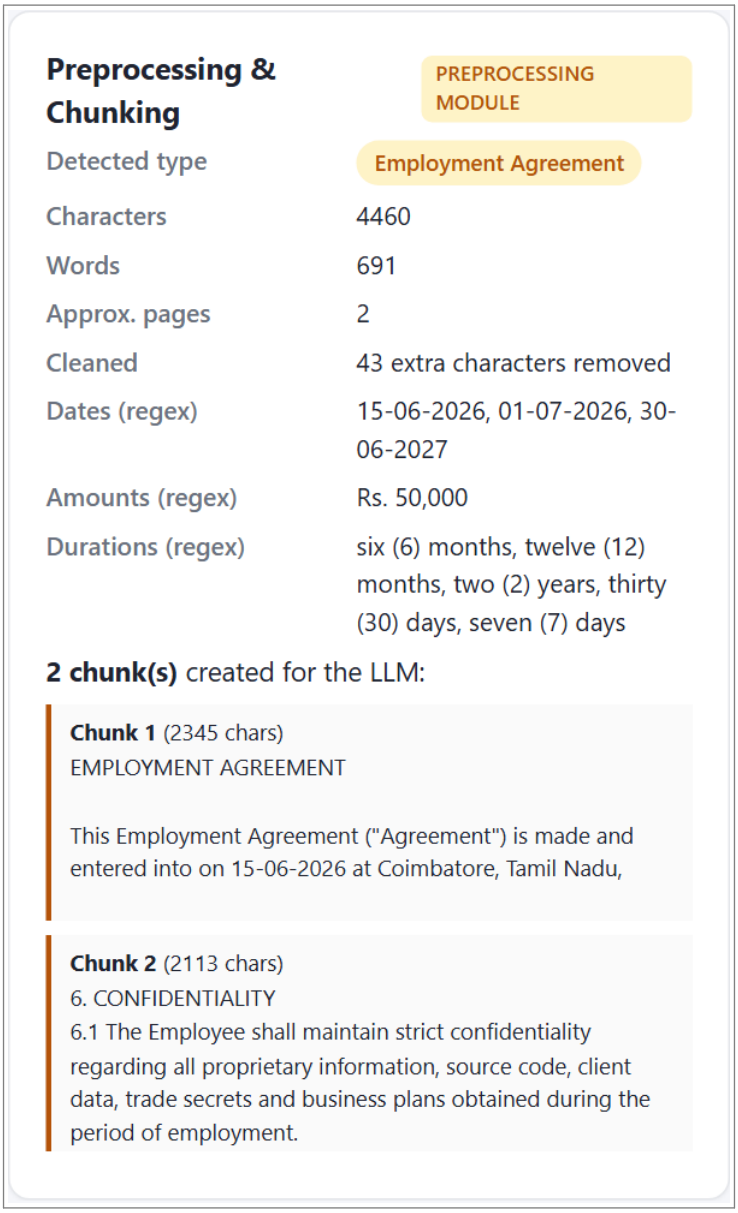


Fig. 5.5 Preprocessing And Chunking

Prompt Generation PROMPT MODULE

Prompt sent for chunk 1 (each chunk gets the same structure).

```
ROLE: You are a legal document analysis assistant.
TASK: Read part 1 of 2 of a Employment Agreement and extract key information.
CONSTRAINT: Use ONLY information present in this text. Do not invent facts. Use [] or "" if not present.
RULES:
- important_dates: include every calendar date that is PRINTED in the text (such as 01-07-2026), copied exactly, with
what happens on it. Never calculate, guess or infer a date that is not printed - a probation or renewal date you worked
out yourself must NOT be listed. A notice period such as "30 days" is NOT a date: put it in termination, not here.
- governing_law: the law or country the contract is governed by, for example "the laws of India". Look for the words
"governed by".
- parties: give the actual names only, not role words like "Employee" on their own.
- obligations: attribute each duty to the party named as the DOER of that sentence, never to the other side. Check the
subject of the sentence before you answer:
    "the Employee shall devote full working time" -> the Employee's duty.
    "the Employee hereby assigns to the Employer all rights in intellectual property" -> the Employee's duty (the
Employee gives, the Employer receives).
    "the Employer shall pay a gross salary" -> the Employer's duty.
  Inside the obligation text keep the role words exactly as the document writes them ("to the Employer", "by the
Employee"). Never swap them round and never replace a role word with a person's or company's name.
  Do not list the same duty twice.
OUTPUT FORMAT: JSON with keys:
{"parties": ["..."], "purpose": "", "duration": "", "payment_terms": ["..."], "obligations": [{"party": "...",
"obligation": "..."}], "termination": ["..."], "confidentiality": "", "governing_law": "", "important_dates": [{"date":
"...", "event": "..."}]}

TEXT:
EMPLOYMENT AGREEMENT

This Employment Agreement ("Agreement") is made and entered into on 15-06-2026 at Coimbatore, Tamil Nadu,

BETWEEN

Nimbus Softech Private Limited, a company incorporated under the Companies Act, 2013, having its registered office at
```

Fig. 5.6 Prompt Generation

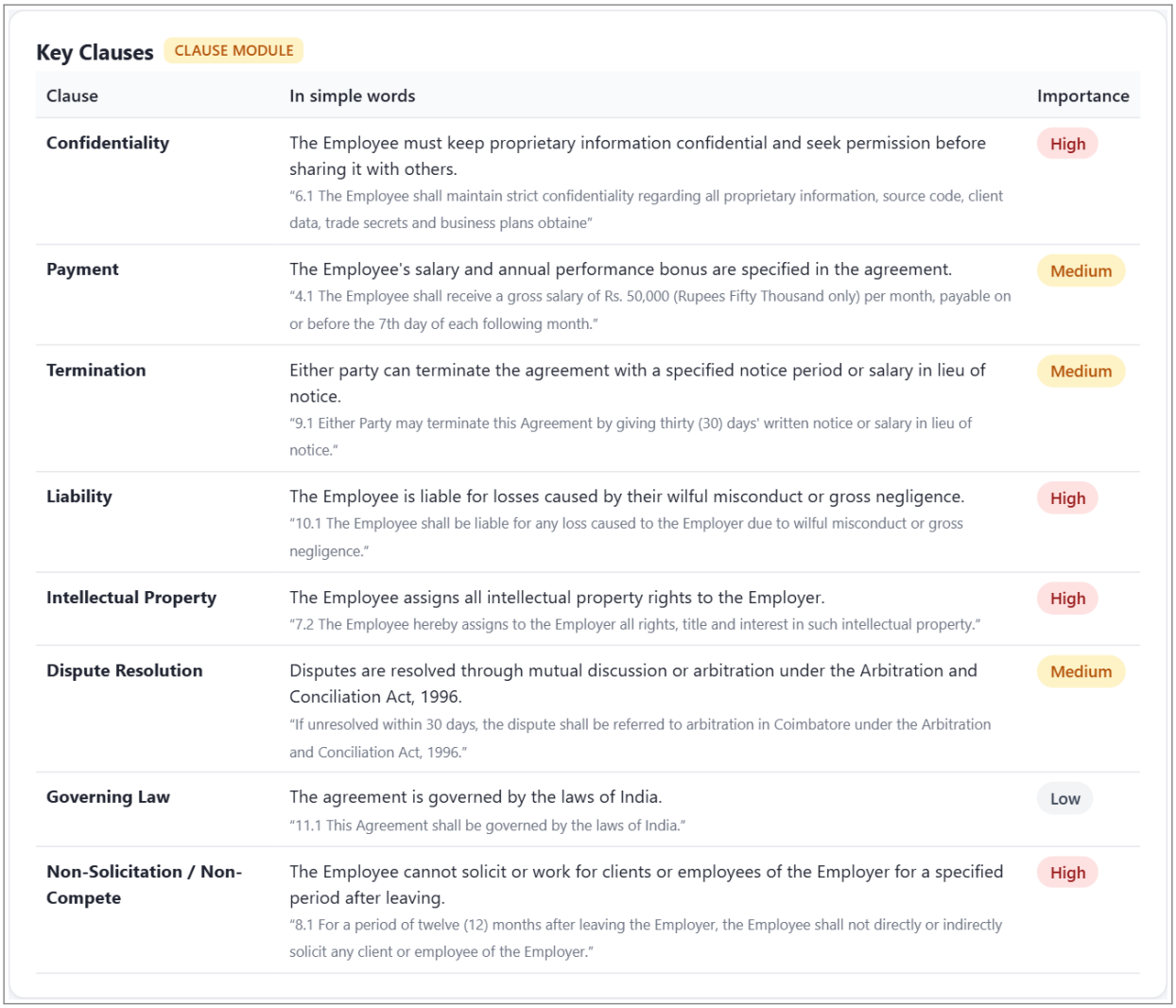
Document Summary SUMMARY MODULE

This Employment Agreement is between Nimbus Softech Private Limited and Mr. Karthik Raman, outlining the terms of Mr. Karthik Raman's employment with the company. The agreement will last for 12 months, starting from July 1, 2026, and can be terminated by either party with 30 days' written notice or by paying the employee's salary in lieu of notice. During this period, Mr. Karthik Raman will perform various duties, including software development and client support, and will maintain confidentiality regarding the company's proprietary information. Upon termination, Mr. Karthik Raman will return all company property and may be liable for any loss caused to the company due to his actions. The agreement is governed by the laws of India and will expire on June 30, 2027.

Points to check before signing:

- The agreement can be terminated by either party with 30 days' written notice or by paying the employee's salary in lieu of notice.
- Mr. Karthik Raman is required to return all company property upon termination.
- The agreement may be terminated early if Mr. Karthik Raman engages in misconduct, fraud, or wilful negligence.

Fig. 5.7 Document Summary



Key Clauses CLAUSE MODULE

Clause	In simple words	Importance
Confidentiality	The Employee must keep proprietary information confidential and seek permission before sharing it with others. "6.1 The Employee shall maintain strict confidentiality regarding all proprietary information, source code, client data, trade secrets and business plans obtaine"	High
Payment	The Employee's salary and annual performance bonus are specified in the agreement. "4.1 The Employee shall receive a gross salary of Rs. 50,000 (Rupees Fifty Thousand only) per month, payable on or before the 7th day of each following month."	Medium
Termination	Either party can terminate the agreement with a specified notice period or salary in lieu of notice. "9.1 Either Party may terminate this Agreement by giving thirty (30) days' written notice or salary in lieu of notice."	Medium
Liability	The Employee is liable for losses caused by their wilful misconduct or gross negligence. "10.1 The Employee shall be liable for any loss caused to the Employer due to wilful misconduct or gross negligence."	High
Intellectual Property	The Employee assigns all intellectual property rights to the Employer. "7.2 The Employee hereby assigns to the Employer all rights, title and interest in such intellectual property."	High
Dispute Resolution	Disputes are resolved through mutual discussion or arbitration under the Arbitration and Conciliation Act, 1996. "If unresolved within 30 days, the dispute shall be referred to arbitration in Coimbatore under the Arbitration and Conciliation Act, 1996."	Medium
Governing Law	The agreement is governed by the laws of India. "11.1 This Agreement shall be governed by the laws of India."	Low
Non-Solicitation / Non-Compete	The Employee cannot solicit or work for clients or employees of the Employer for a specified period after leaving. "8.1 For a period of twelve (12) months after leaving the Employer, the Employee shall not directly or indirectly solicit any client or employee of the Employer."	High

Fig. 5.8 Key Clauses

Obligations, Payment & Dates ANALYSIS MODULE

Obligations

Party	Obligation
Mr. Karthik Raman	The Employee shall perform all duties assigned by the reporting manager, including software development, testing, documentation and client support.
Mr. Karthik Raman	The Employee shall maintain strict confidentiality regarding all proprietary information, source code, client data, trade secrets and business plans obtained during the period of employment.
Mr. Karthik Raman	The Employee shall not disclose such information to any third party without prior written authorization from the Employer, both during employment and for a period of two (2) years after termination.
Mr. Karthik Raman	All software, inventions, designs, documents and other work created by the Employee in the course of employment shall be the sole and exclusive property of the Employer.
Mr. Karthik Raman	The Employee hereby assigns to the Employer all rights, title and interest in such intellectual property.
Mr. Karthik Raman	For a period of twelve (12) months after leaving the Employer, the Employee shall not directly or indirectly solicit any client or employee of the Employer.
Nimbus Softech Private Limited	Either Party may terminate this Agreement by giving thirty (30) days' written notice or salary in lieu of notice.
Employee	During the probation period, either Party may terminate this Agreement with seven (7) days' written notice.
Nimbus Softech Private Limited	The Employer may terminate employment immediately without notice in case of misconduct, fraud, breach of confidentiality or wilful negligence.
Mr. Karthik Raman	On termination, the Employee shall return all company property, including laptop, ID card and documents.
Mr. Karthik Raman	The Employee shall be liable for any loss caused to the Employer due to wilful misconduct or gross negligence.

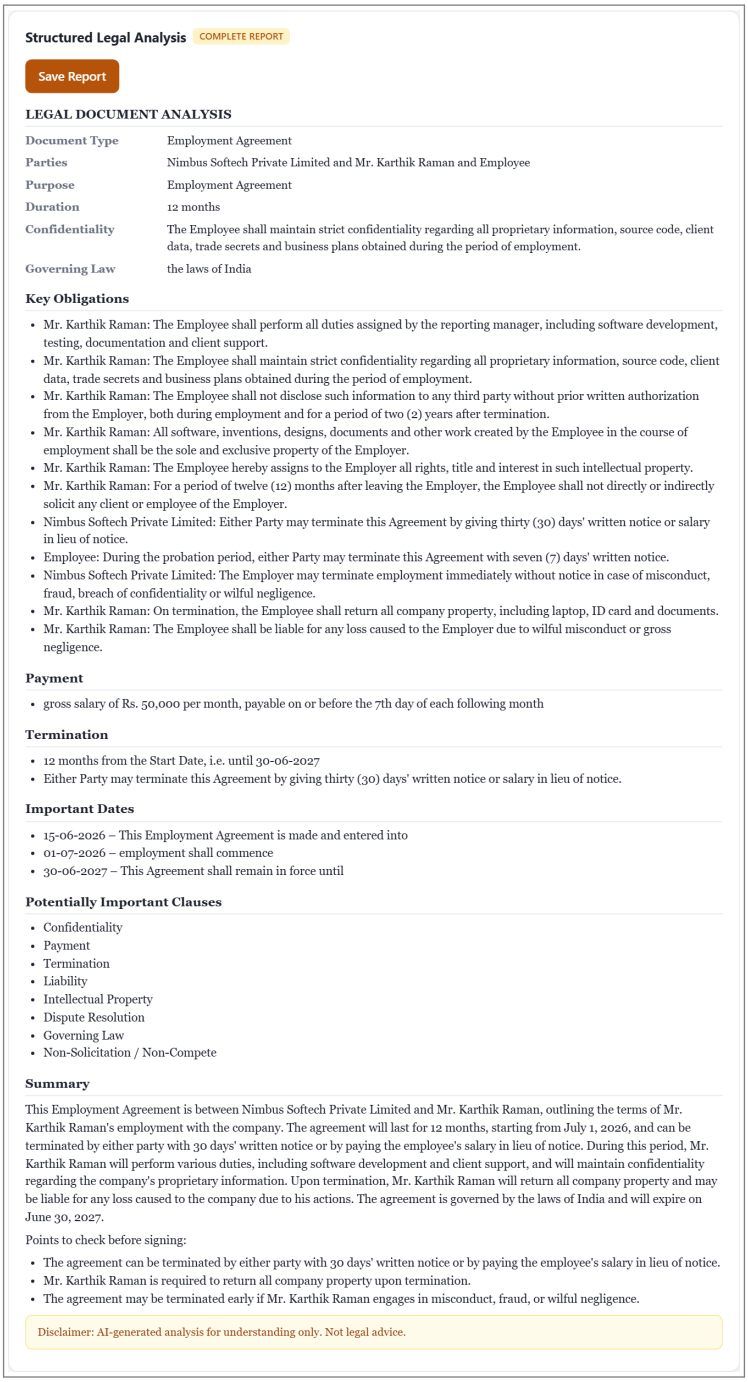
Payment terms

- gross salary of Rs. 50,000 per month, payable on or before the 7th day of each following month

Important dates

Date	Event
15-06-2026	This Employment Agreement is made and entered into
01-07-2026	employment shall commence
30-06-2027	This Agreement shall remain in force until

Fig. 5.9 Obligations Payment And Dates



Structured Legal Analysis COMPLETE REPORT

Save Report

LEGAL DOCUMENT ANALYSIS

Document Type	Employment Agreement
Parties	Nimbus Softech Private Limited and Mr. Karthik Raman and Employee
Purpose	Employment Agreement
Duration	12 months
Confidentiality	The Employee shall maintain strict confidentiality regarding all proprietary information, source code, client data, trade secrets and business plans obtained during the period of employment.
Governing Law	the laws of India

Key Obligations

- Mr. Karthik Raman: The Employee shall perform all duties assigned by the reporting manager, including software development, testing, documentation and client support.
- Mr. Karthik Raman: The Employee shall maintain strict confidentiality regarding all proprietary information, source code, client data, trade secrets and business plans obtained during the period of employment.
- Mr. Karthik Raman: The Employee shall not disclose such information to any third party without prior written authorization from the Employer, both during employment and for a period of two (2) years after termination.
- Mr. Karthik Raman: All software, inventions, designs, documents and other work created by the Employee in the course of employment shall be the sole and exclusive property of the Employer.
- Mr. Karthik Raman: The Employee hereby assigns to the Employer all rights, title and interest in such intellectual property.
- Mr. Karthik Raman: For a period of twelve (12) months after leaving the Employer, the Employee shall not directly or indirectly solicit any client or employee of the Employer.
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- Nimbus Softech Private Limited: The Employer may terminate employment immediately without notice in case of misconduct, fraud, breach of confidentiality or wilful negligence.
- Mr. Karthik Raman: On termination, the Employee shall return all company property, including laptop, ID card and documents.
- Mr. Karthik Raman: The Employee shall be liable for any loss caused to the Employer due to wilful misconduct or gross negligence.

Payment

- gross salary of Rs. 50,000 per month, payable on or before the 7th day of each following month

Termination

- 12 months from the Start Date, i.e. until 30-06-2027
- Either Party may terminate this Agreement by giving thirty (30) days' written notice or salary in lieu of notice.

Important Dates

- 15-06-2026 – This Employment Agreement is made and entered into
- 01-07-2026 – employment shall commence
- 30-06-2027 – This Agreement shall remain in force until

Potentially Important Clauses

- Confidentiality
- Payment
- Termination
- Liability
- Intellectual Property
- Dispute Resolution
- Governing Law
- Non-Solicitation / Non-Compete

Summary

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- Mr. Karthik Raman is required to return all company property upon termination.
- The agreement may be terminated early if Mr. Karthik Raman engages in misconduct, fraud, or wilful negligence.

Disclaimer: AI-generated analysis for understanding only. Not legal advice.

Fig. 5.10 Complete Structured Analysis



Report Saved SAVE MODULE

✓ Analysis report saved successfully

File E:\gpt4\GPT4-Lab-URK23CS1167\GPT4-Exp5-Legal-Document-Analyzer\saved\analysis_employment_agreement_20261102_110634.txt

Size 5197 bytes

Fig. 5.11 Report Saved

Result:

Thus, an automated legal document analyzer using GPT-4 was successfully developed. The system accepts legal documents, processes the text, generates a concise summary, identifies important clauses and obligations, extracts relevant information and produces a structured analysis.